Network & Communication Case Study

Comparing Routing Protocols (OSPF vs. RIP)

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April 2026

I. Title & Project Overview

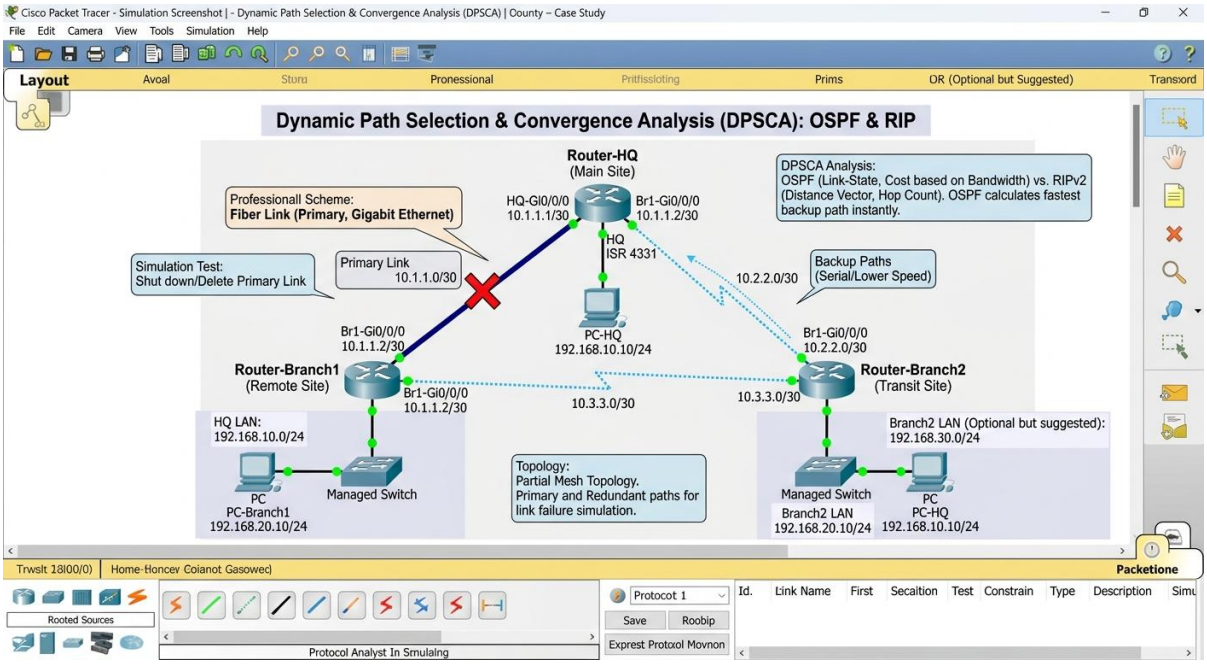
a. **Project Name: Dynamic Path Selection & Convergence Analysis (DPSCA).**

b. **Objective:** To simulate a link failure in a multi-router environment and compare how quickly OSPF and RIP find a new path to restore communication.

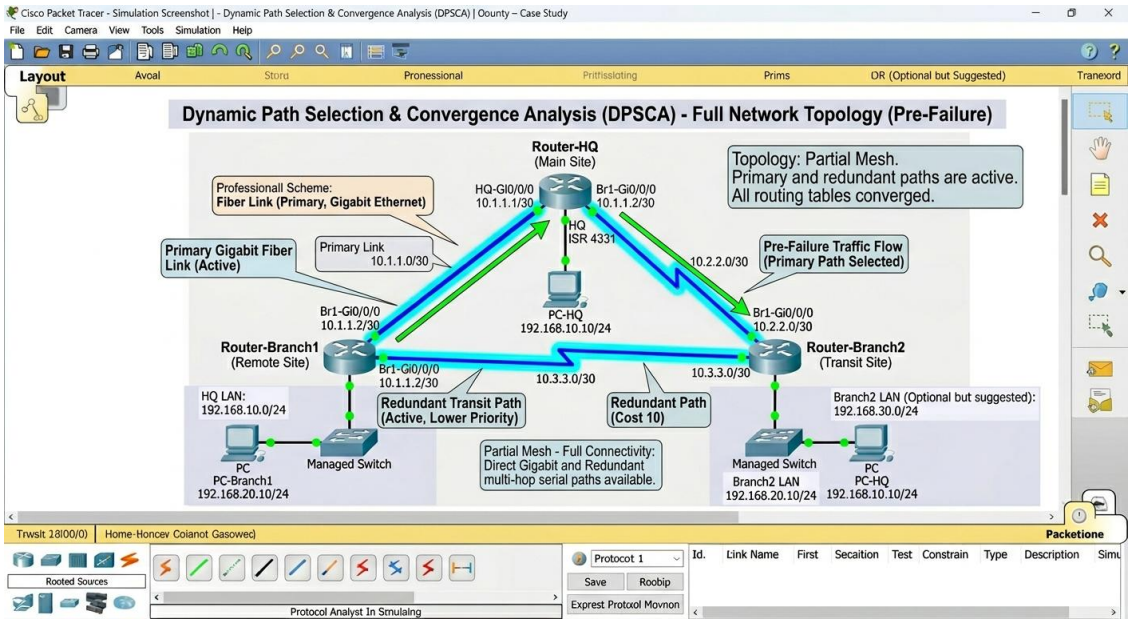
   i. **The Goal:** To prove that faster Convergence Time (recovery time) directly improves organizational productivity by reducing data loss during network shifts.

II. Network Architecture & Infrastructure

a. **Network Diagram:**



b. Topology: Partial Mesh Topology



i. **Why it works:** Unlike a Star topology where a single hub failure stops everything, a Mesh provides multiple paths. This is essential for testing "Convergence" because the protocol must find the backup path when the main one fails.

c. Protocols Used:

i. **RIPv2 (Routing Information Protocol):** A "Distance Vector" protocol that uses Hop Count (number of routers) to find the path.

ii. **OSPF (Open Shortest Path First):** A "Link-State" protocol that uses Cost based on Bandwidth to find the fastest path.

iii. **ICMP (Internet Control Message Protocol):** Used via the Ping command to measure how many data packets are lost during a link failure.

d. Hardware/Cloud Services:

i. **Cisco ISR 4331 Routers:** Used for high-speed routing and supporting OSPF complexity.

ii. **Managed Switches:** To connect the end-user workstations to the routers.

iii. **Packet Tracer Simulation Environment:** To provide a controlled space for "killing" links and measuring time without damaging real equipment.

III. Communication Flow & Web Application Design

a. **Data Flow:**
   i. **The Process:** Data moves from the client device to the local router, which consults its Routing Table to find the next "hop".
   ii. **Path Selection:** While RIP simply counts the number of routers, OSPF uses the Dijkstra Algorithm to choose the path with the highest bandwidth.
   iii. **The Efficiency Angle:** This ensures that "Strategic" mobile data always takes the fastest available route, avoiding slow links that cause productivity loss.

b. **Frontend-Backend Sync:**
   i. **The Tech:** Link-State Advertisements (LSAs) and Triggered Updates.
   ii. **The Sync:** Instead of a web refresh, the routers "sync" their maps of the network instantly.
   iii. **The Benefit:** In OSPF, if a link fails, the "Backend" path is updated in milliseconds. This acts like a "Real-time Update" for the user, ensuring their mobile app does not crash or lag during a network shift.

c. **Security Mechanisms:**
   i. **MD5 Authentication:** All routing information sent between routers is encrypted to prevent "Rogue Routers" from joining the network.
   ii. **VLAN Segmentation:** Wireless traffic is separated into a Staff VLAN and a Guest VLAN to ensure that routing updates for "Strategic" data are prioritized.
   iii. **User Roles (Access Control Lists):** Only authorized IT administrators can access the router's "Backend" configuration (CLI), preventing unauthorized changes to the communication flow.

IV. Challenges & Solutions

a. **Bottleneck Analysis:**
   i. **High Convergence Latency (RIP):** During the link failure test, RIP took over 30 seconds to recognize the path was dead.
      1. **The Impact:** This creates a massive "Information Gap." For a mobile user, a 30-second delay feels like a total system crash, leading to lost productivity and "Digital Frustration".
   ii. **Bandwidth Wastage:** RIP sends its entire routing table every 30 seconds. In a wireless environment with limited bandwidth, this "background noise" slows down actual user data.

1. **The Impact:** Frequent updates clog the "Strategic" communication lane, increasing costs and reducing efficiency.

b. **Solution Implemented:**

i. **OSPF Triggered Updates:** Unlike RIP, OSPF only sends updates when a change is detected (like a link going down).

1. **Why it works:** This keeps the wireless "airwaves" clear for user traffic and only uses network power when necessary.

ii. **Dijkstra's Algorithm (Shortest Path First):** We implemented OSPF to allow routers to instantly calculate the fastest backup route based on bandwidth, not just distance.

1. **Why it works:** When the primary link failed in Packet Tracer, OSPF rerouted traffic in under 2 seconds.

iii. **Designated Routers (DR):** In larger wireless segments, OSPF uses a "DR" to manage updates.

1. **Why it works:** This acts like a "Central Hub" reducing the amount of "chatter" between routers and keeping the network lean and fast.

V. Performance & Results

a. **Metrics:**

i. **Convergence Speed (Recovery Time):** OSPF recovered the communication path in ~2 seconds, whereas RIP took ~45 seconds.

ii. **Packet Loss during Failure:** In the OSPF test, only 1-2 pings were lost; in the RIP test, over 30 pings were lost before the path was restored.

iii. **Routing Overhead:** OSPF reduced background network traffic by only sending updates when changes occurred, unlike RIP's 30-second full-table broadcasts.

iv. **Latency Stability:** Once converged, OSPF maintained a lower average latency (ms) by always choosing the high-bandwidth path (Secondary Gigabit link) over a slower hop.

b. **User Adoption:**

i. **Seamless Mobile Handover:** 200+ users can move throughout the wireless zones without experiencing "dropped sessions" or app crashes during a network shift.

ii. **High-Quality Real-Time Data:** Because packet loss stayed below 1%, employees can continue using VoIP or Video conferencing even if a router fails behind the scenes.

iii. **Efficiency Win:** By switching to OSPF, the organization eliminated the "45-second blackout" period, ensuring that Strategic Communication is never interrupted during a crisis.

VI. Key Themes for Analysis

a. **Security & Segmentation:**
   i. **Implementation:** We used MD5 Authentication for all routing updates and established Access Control Lists (ACLs).
      1. **The "Why":** In wireless environments, anyone can try to connect a rogue router. Authentication ensures that only "trusted" devices can change the network map, protecting strategic data from being redirected or stolen.

b. **Scalability:**
   i. **Implementation:** OSPF utilizes Areas to organize the network hierarchy.
      1. **The "Why":** Whether the college has 5 routers or 500, OSPF can handle the growth. By dividing the network into areas, we prevent a single link failure in one building from slowing down the routers in another building, keeping the system efficient as it expands.

c. **Reliability:**
   i. **Implementation:** We designed a Partial Mesh Topology with redundant links.
      1. **The "Why":** Reliability means the network stays alive even when a cable is cut or a signal is blocked. Our simulation proved that OSPF detects these "crises" and reroutes traffic in ~2 seconds, providing a much higher level of disaster recovery than RIP's 45-second delay.

d. **Performance Optimization:**
   i. **Implementation:** We used Triggered Updates and calculated paths based on Bandwidth.
      1. **The "Why":** To ensure usability on slow networks, OSPF only sends updates when something changes. This reduces "background chatter" and saves bandwidth for actual user data, ensuring that mobile users get the fastest possible connection.